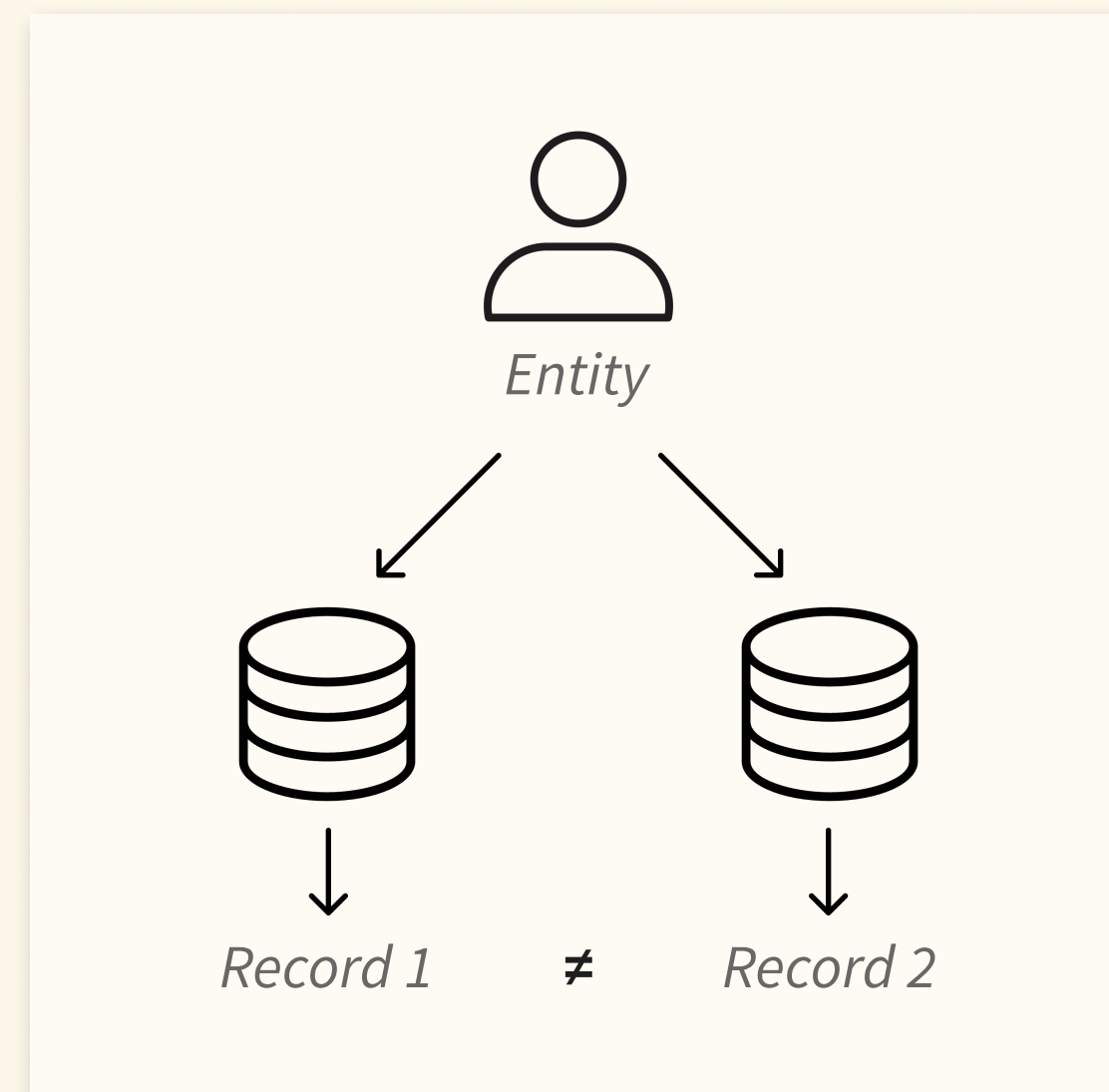


Chapter 3.8

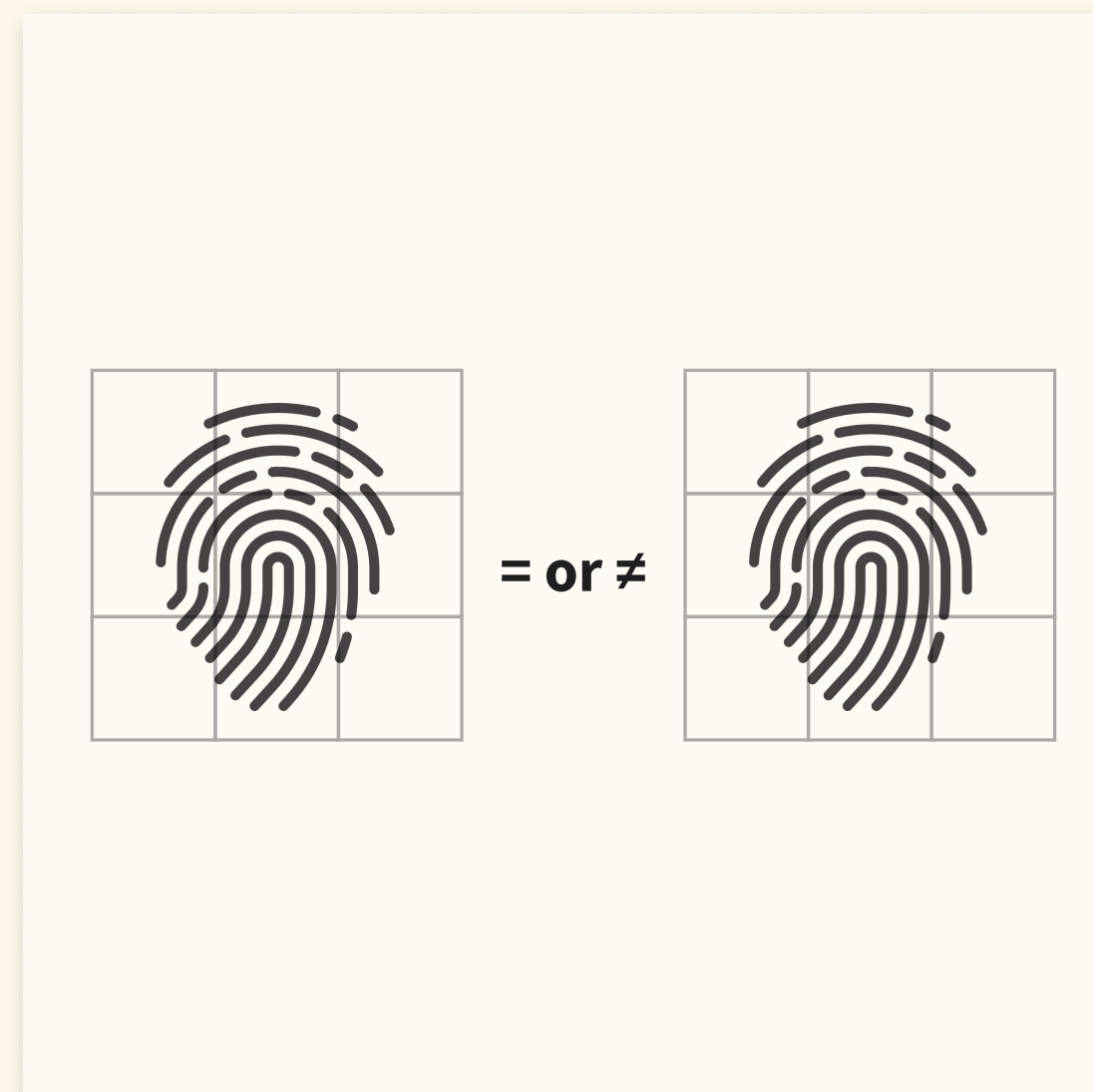
Applications of Locality-Sensitive Hashing

Chapter 3.8

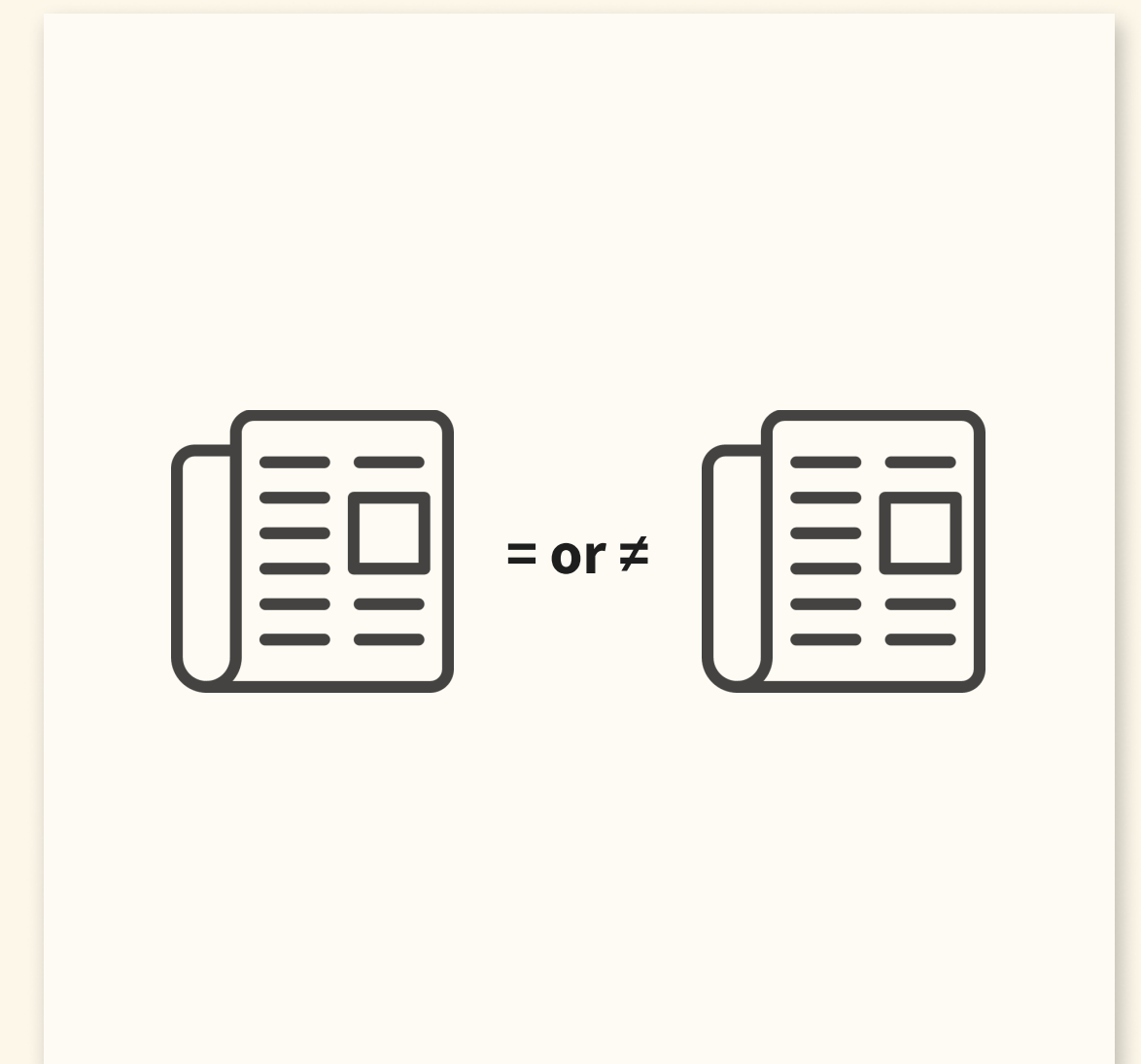
Applications of Locality-Sensitive Hashing



Entity Resolution



Matching Fingerprints



Matching Newspaper Articles

Chapter 3.8

Applications of Locality-Sensitive Hashing

1. Entity Resolution
2. An Entity-Resolution Example
3. Validating Record Matches
4. Matching Fingerprints
5. A LSH Family for Fingerprint Matching
6. Similar News Articles

3.8.1

Entity Resolution

Entity Resolution

The problem of matching different datasets when they represent the same entity

Why do different datasets exist for the same entity?

1. Misspellings, absence of a middle initial, use of a nickname, and many other reasons

ex) “*Bob S. Jomes*” and “*Robert Jones Jr.*”

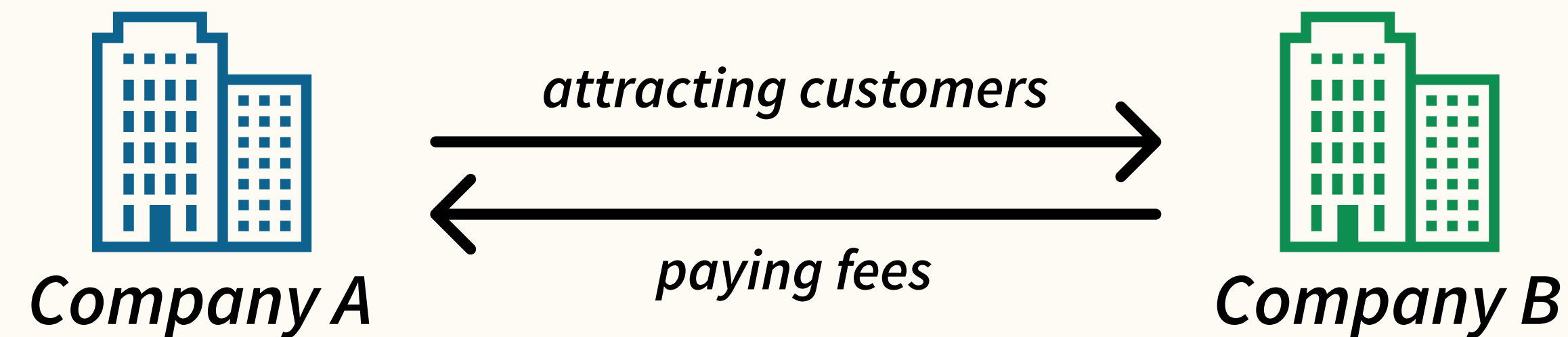
2. Different fields collected from different sources

ex) “*age*” field and “*date of birth*” field

3.8.2

An Entity-Resolution Example

2. AN ENTITY-RESOLUTION EXAMPLE



How many customers A had provied to B?

Need to match the two sets of records to verify that the pairs represent the same person

Cause of the problem: **Inconsistencies in the data**



Records A

home phone
address before moving
...



Records B

cell phone
address after moving
...

Fields
Name, Address, Phone

Solve the problem: **Scoring the degree of match in three fields**

- Each field: **100 points**
- If all three fields match: **300 points**
- If mismatches in each field: **Penalized based on edit-distance between two records**
 - Reduce penalties for specific situations (e.g., “Bill” and “William”) using publicly available tables

Realistically, it's impossible to compare all 1 trillion (1 million^2) pairs of records

→ **Use LSH to narrow down likely candidate pairs**

Solve the problem: **Scoring the degree of match in three fields**

Three hash functions:

1. a function that puts records with the same “*name*” into the same bucket
2. a function that puts records with the same “*address*” into the same bucket
3. a function that puts records with the same “*phone*” into the same bucket

Instead of performing hashing, **sorting** was used:

1. Records are sorted by “*name*”, and calculated the similarity of records with the same name
2. Records are sorted by “*address*”, and calculated the similarity of records with the same name
3. Records are sorted by “*phone*”, and calculated the similarity of records with the same name

Limitation: Records that don't match any of the three fields are excluded from the comparison

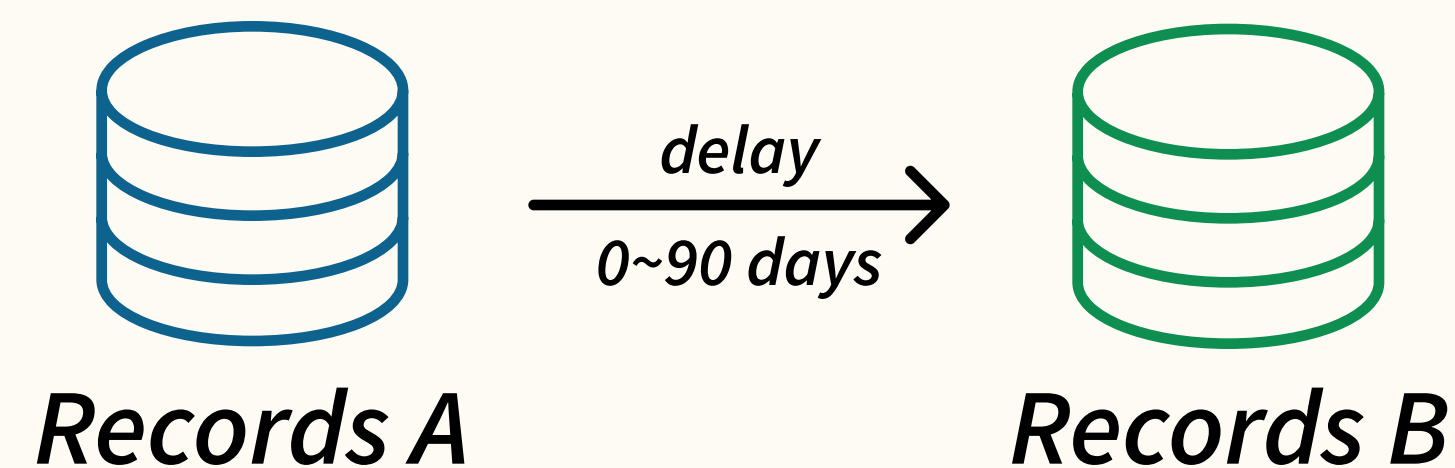
3.8.3

Validating Record Matches

3. VALIDATING RECORD MATCHES

- In Section 3.8.2, a **score** was calculated for the matching of records
- A **threshold score** must be determined to decide whether two records truly represent the same entity

Fields the creation date of the records



- The average delay for **randomly chosen pairs** of records = **45 days**
- The average delay for pairs of records with a **perfect score of 300** = **10 days**
- Let x be **the average delay for pairs with score s**
- Let f be **the fraction of the pairs with score s that are true matches**
= the proportion of pairs whose average delay is 10 days

$$x = 10f + 45(1 - f) = 45 - 35f$$

$$f = (45 - x)/35$$

Generalization

This approach can be applied in the following conditions:

1. There is **a scoring system** used to evaluate the likelihood that two records represent the same entity
2. There is some **field, not used in the scoring**, from which we can derive a **measure that differs, on average, for true pairs and false pairs**

Key Idea

Use **additional information** that is independent of the scoring system to validate accuracy of matches

3.8.4

Matching Fingerprints

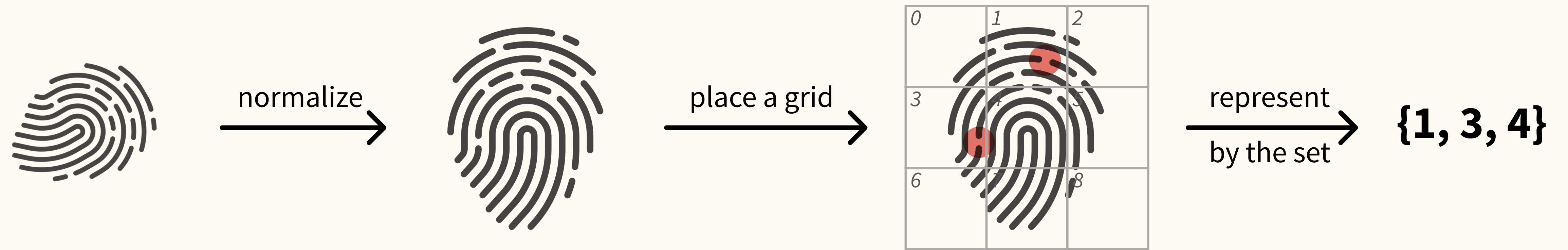
4. MATCHING FINGERPRINTS

Fingerprints Minutiae

When fingerprints are matched by computer, the usual representation is not an image, but a set of locations in which *minutiae* are located



Fingerprints with Grid



Grid squares from two images of same finger have a significantly higher probability of agreeing in the presence or absence of a minutia

Fingerprint Comparison

Step 1. Represent fingerprints by sets of grid squares

Step 2. Compare sets using the Jaccard similarity or distance

Two versions of fingerprint comparison:

1. *Many-to-one*

- The task of comparing one fingerprint (e.g., found on a gun) with all fingerprints in a large database to find a match

2. *Many-to-many*

- The task of finding pairs of fingerprints in the entire database that belong to the same individual

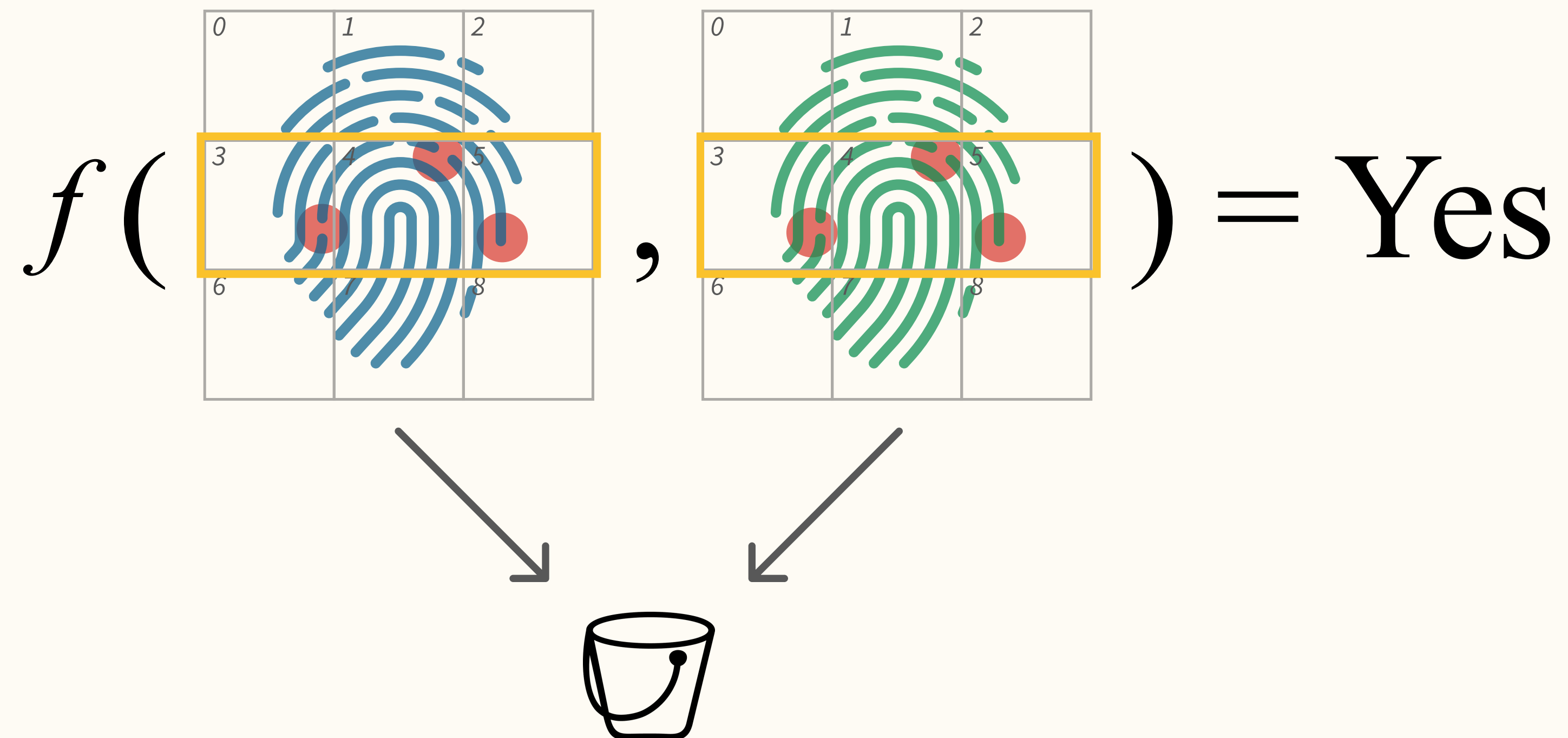
→ Use LSH for efficient comparisons

3.8.5

A LSH Family for Fingerprint Matching

LSH function family F

- Each hash function $f \in F$ defined by **3 specific grid squares**
 - Fingerprints placed in the **same bucket** if they share *minutiae* in **all 3 grid squares**



LSH function family F

- Probability of finding a minutia in **a random grid square of a random fingerprint** = **20%**
- Probability that if **two fingerprints come from the same finger** and **one has a minutia in a given grid square**, the **other will also have a minutia** = **80%**

- Probability of **two fingerprints** from **different fingers** being in the **same bucket** for a function f
= Probability of having minutia in all 3 grid squares of both fingerprints

$$(0.2)^3 \times (0.2)^3 = (0.2)^6 = 0.000064$$

- Probability of **two fingerprints** from **same finger** being in the **same bucket** for a function f
= Probability of having minutia in 3 grid squares $\times$ Probability of having minutia in 3 grid squares of other fingerprints

$$(0.2)^3 \times (0.8)^3 = 0.004096$$

Improve Match Probability with **OR Construction**

Step 1. Use 1,024 functions from F

Step 2. Construction F_I by performing
a 1024-way **OR** operation

OR operation

$(d_1, d_2, 1 - (1 - p_1)^b, 1 - (1 - p_2)^b)$ - sensitive family

- Probability that F_I will put fingerprints from the **same finger** together in at least one bucket

$$1 - (1 - 0.004096)^{1024} = 0.985$$



1.5% False Negatives

- Probability that F_I will put fingerprints from the **different finger** together in at least one bucket

$$(1 - (1 - 0.000064)^{1024}) = 0.063$$



6.3% False Positives

Improve Match Probability with **AND Construction**

Step 1. Use 2048 functions from F in two groups of 1024

Step 2. In each group of functions, compute the **union** of the buckets that a given fingerprint P belongs to

Step 3. Calculate the intersection of two union sets

Step 4. Compare P with the fingerprints in this intersection

AND operation

$(d_1, d_2, (p_1)^r, (p_2)^r)$ - sensitive family

- Probability that F_I will put fingerprints from the **same finger** together in at least one bucket

$$(0.985)^2 = 0.970$$



3% False Negatives

- Probability that F_I will put fingerprints from the **different finger** together in at least one bucket

$$(0.063)^2 = 0.00397$$



0.397% False Positives

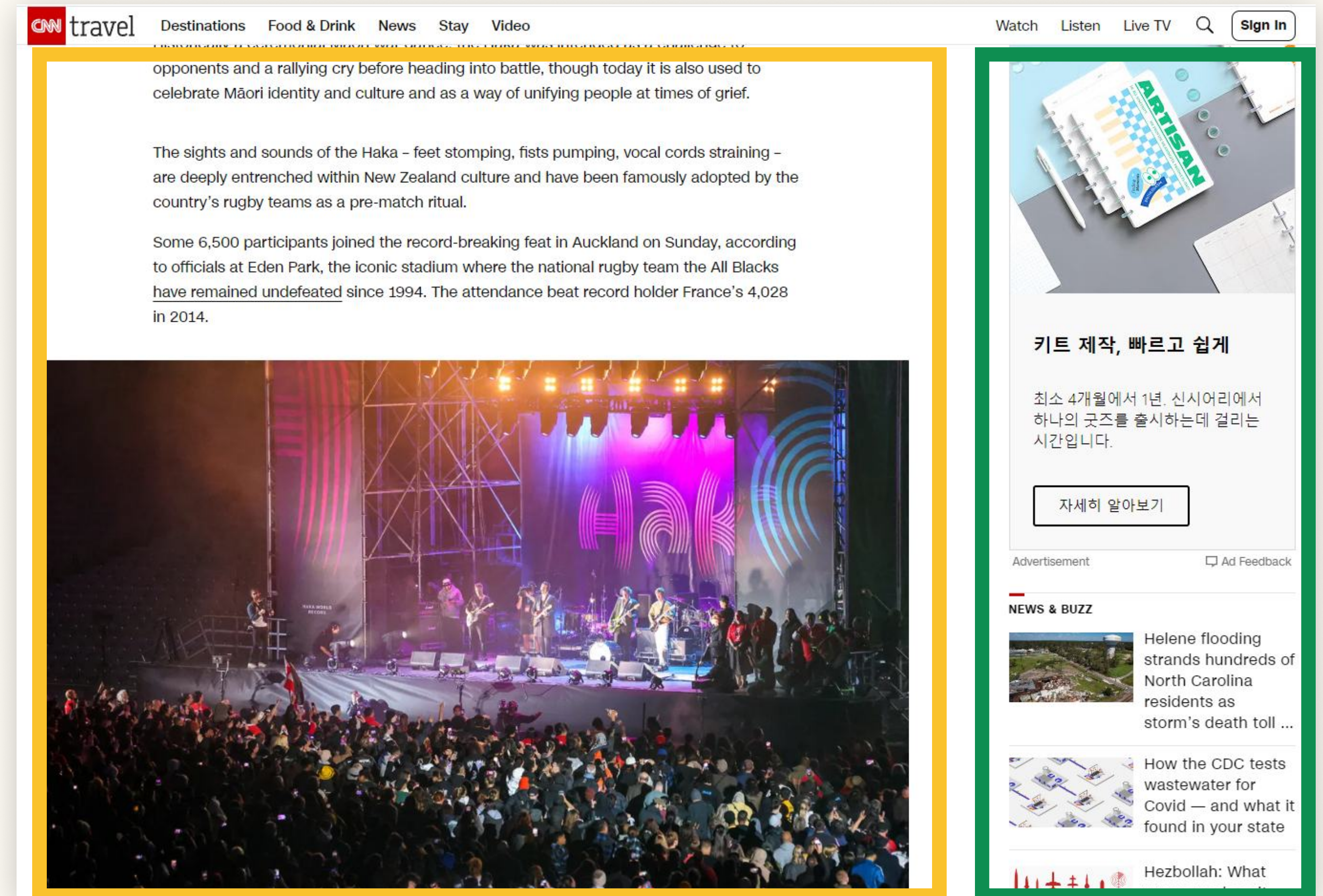
3.8.6

Similar News Articles

6. SIMILAR NEWS ARTICLES

News

- **The body of a news article:** stop words like “the,” “and”, etc. occur frequently.
- **Surrounding elements** such as ads: stop words are not frequent



The body of a news article

Surrounding elements

Apply **Shingling**

- **Definition of shingles:** a stop word followed by the next two words
- **Ad:** “*Buy Sudzo*”
 - Shingles: None
- **News article:** “*I recommend that you buy Sudzo for your laundry*”
 - Shingles
 - i. “*I recommend that*”
 - ii. “*that you buy*”
 - iii. “*you buy Sudzo*”
 - iv. “*for your laundry*”
 - v. “*your laundry x*”



Shingles of Total page:

0 shingle from Ad + 5 shingles from news article

Thank you